Cold-atom comagnetometry via optical control of spin states

J.-L. Zhang[1†], W.-T. Luo[1†], Y. A. Yang[1], Y.-Q. Wang[1], T. Xia[2*], Z.-T. Lu[1,2*]

[1]Hefei National Research Center for Physical Sciences at the Microscale, School of Physical Sciences, University of Science and Technology of China, Hefei, Anhui, 230026, China.
[2]Hefei National Laboratory, University of Science and Technology of China, Hefei, Anhui, 230088, China.

*Corresponding author(s). E-mail(s): txia@hfnl.cn; ztlu@ustc.edu.cn;
Contributing authors: jielin@mail.ustc.edu.cn; luowt@mail.ustc.edu.cn; yyustcer@mail.ustc.edu.cn; wyq0705@mail.ustc.edu.cn;
†These authors contributed equally to this work.

Abstract

Atomic spin-based comagnetometers are powerful tools for precision sensing and tests of fundamental physics. Compared with the widely used gas-cell comagnetometer systems, cold-atom systems offer access to much shorter distance scales and allow implementation of optical quantum control techniques. However, in order to realize long spin coherence times with cold atoms, it is necessary to employ diamagnetic atoms and overcome decoherence induced by light shifts. Here we demonstrate a cold-atom comagnetometer based on the nuclear spins of ^{171}Yb (spin-1/2) and ^{173}Yb (spin-5/2), jointly trapped in an optical lattice. Vector light shifts are suppressed by enforcing linear polarization of the lattice, while tensor shifts in ^{173}Yb are suppressed via the use of a Schrödinger cat state. This enables simultaneous Ramsey interferometry on both isotopes with a spin coherence time of 60 s. We achieve a magnetic noise suppression factor exceeding $\mathbf{3 \times 10^4}$, and determine the ratio of nuclear magnetic moments as -0.726076(3), with a relative uncertainty of 4 ppm. ~~to 4 ppm precision.~~ Our results establish a new cold-atom platform

for spin-based sensing and open pathways toward quantum-enhanced searches for physics beyond the Standard Model.

Keywords: optical quantum control, Ytterbium, comagnetometer, suppression of magnetic noise, spin sensor, nuclear magnetic moment

Introduction

Atomic-spin-based sensors are indispensable in quantum information science [1] and precision measurement [2, 3]. Paramagnetic atoms, whose spins carry Bohr magnetons, underpin some of the most sensitive magnetometers [4]. In contrast, diamagnetic atoms possess spins originating solely from their nuclei. These nuclear spins, governed by much smaller nuclear magnetons, exhibit significantly longer coherence times [5], making them ideal for sensing interactions beyond magnetism. Nuclear magnetic resonance (NMR) is a prominent example of such sensing.

In order to suppress magnetic-field noise, a comagnetometer employs two atomic species with distinct gyromagnetic ratios, colocated and measured simultaneously [6]. The ratio of their Larmor frequencies ~~—termed the comagnetometer ratio—~~ is intrinsically insensitive to magnetic-field fluctuations while retaining sensitivity to nonmagnetic interactions. Such systems have demonstrated the exceptional ability to detect energy shifts at the pHz level [7].

Room-temperature gas-cell comagnetometers have been extensively developed, with combinations including the ^{85}Rb-^{87}Rb paramagnetic pair [8], paramagnetic-diamagnetic pairs K-^{3}He [7] and Rb-^{21}Ne [9], and diamagnetic pairs ^{3}He-^{129}Xe [9–11], Xe isotopes [12, 13] and Hg isotopes [14]. These platforms support a wide range of applications, from inertial sensing [15] to precision tests of fundamental physics, including searches for electric dipole moments (EDMs) [16–18], monopole–dipole interactions [12, 13, 19], spin–gravity couplings [7, 8, 20, 21], and axion-like dark matter [22, 23]. However, their centimeter-scale cell sizes and millimeter-thick walls impose a limitation on sensitivity to short-range interactions with ranges below $\sim 100\ \mu$m.

Cold-atom comagnetometers can overcome this limitation. Atom clouds confined in a laser trap can be reduced to scales of 1–100 μm, enabling sensitivity to comparable interaction ranges. Moreover, the absence of material walls eliminates related systematic effects. A prior implementation used two hyperfine states of paramagnetic ^{87}Rb with spin coherence times at several milliseconds [24]. Signals of the two hyperfine states were combined to generate a single-species magnetometer based on the nuclear spin with a coherence time of 1 s. More recently, a cold-atom system combining paramagnetic ^{87}Rb with diamagnetic ^{87}Sr was realized [25]. Operating under magnetic fields of

100-1000 G, this system enabled ppm-level determination of the nuclear g factor of ^{87}Sr, though its coherence time was ultimately limited to milliseconds by the ^{87}Rb component under strong magnetic fields.

Here, we realize a comagnetometer based purely on nuclear spins consisting of ^{171}Yb (spin-1/2) and ^{173}Yb (spin-5/2), simultaneously confined in a one-dimensional optical lattice under a weak magnetic field (10-100 mG). A key challenge in such systems is the suppression of lattice-induced light shifts. We eliminate vector light shifts by enforcing linear polarization of the lattice beam. Tensor shifts are naturally absent in ^{171}Yb due to its spin-1/2 character. For the higher-spin ^{173}Yb, we apply quantum control techniques to prepare a Schrödinger cat state, which is structurally analogous to a NOON state [26] and exhibits a precession frequency rendered insensitive to tensor light shifts. ~~whose precession frequency is rendered insensitive to tensor shifts.~~ Using simultaneous Ramsey interferometry on both isotopes, we measure the Larmor-frequency ratio ~~comagnetometer ratio~~ and demonstrate a spin coherence time of 60 s. We identify a novel systematic effect in the cat state stemming from the interplay between Zeeman and tensor light shifts. By understanding and controlling both vector and tensor light shifts, we achieve a magnetic-noise suppression factor exceeding 3×10^4. We determine the ratio of nuclear magnetic moments between ^{171}Yb and ^{173}Yb with a precision of 4 ppm.

Realization of cold-atom comagnetometer

In preparation of a cold-atom ensemble, ytterbium atoms containing both ^{171}Yb and ^{173}Yb (isotopic abundances: 14% and 16%) exit an oven, pass through a Zeeman slower, and are captured by a blue magneto-optical trap (MOT) formed with a laser exciting the broad transition $6s^2\ {}^1\mathrm{S}_0 \rightarrow 6s6p\ {}^1\mathrm{P}_1$ at 399 nm. The atoms are subsequently cooled to a temperature of $\sim 20\ \mu$K with a green MOT at 556 nm acting on the narrow transition $6s^2\ {}^1\mathrm{S}_0 \rightarrow 6s6p\ {}^3\mathrm{P}_1$, and then transferred to a movable optical dipole trap (Bus ODT) at 1036.4 nm, which is a magic wavelength for the cooling transition [27]. Carried by the Bus ODT from the MOT chamber to an adjacent science chamber across 0.35 m, the atoms arrive at their final destination: a horizontally directed, one-dimensional optical lattice also at 1036.4 nm (Fig. 1a). With a beam power of 30 W and a waist of 25 μm, the lattice consists of trap sites with a depth of ~ 2 mK. A total of about 2×10^4 ^{171}Yb atoms and 2×10^4 ^{173}Yb atoms are mixed together to form an ellipsoidal cloud with a diameter of 40 μm and a length of 80 μm, occupying approximately 160 lattice sites. The lifetime of atoms in the optical lattice is observed to be 50 s.

The science chamber is surrounded by a set of coils and magnetic shields. A uniform magnetic field of 10 - 100 mG in the Z-direction is applied to define the spin quantization axis (Fig. 1a). In the ${}^1\mathrm{S}_0, F = 1/2$ ground state of ^{171}Yb, a transversely polarized state $|\psi_t\rangle = |F = 1/2, m_F = +1/2\rangle_g + \mathrm{e}^{\mathrm{i}\phi_1} |F = 1/2, m_F = -1/2\rangle_g$ is employed to sense the precession phase ϕ_1. For

the ground state of ^{173}Yb (I = 5/2), it is necessary to prepare atoms into the cat state, $|\psi_c\rangle = |5/2, +5/2\rangle_g + e^{i\phi_2} |5/2, -5/2\rangle_g$, in order to achieve immunity to otherwise significant decoherence caused by tensor light shifts in the optical lattice [28]. The cat state cannot simply be prepared with either optical pumping or a rotating magnetic field. Instead, it is generated using a 556 nm Rabi beam, a circularly polarized laser beam traveling in the X-direction to drive the necessary Rabi transitions, with its $\sim$ GHz detunings from the $6s^2\ ^1\mathrm{S}_0 \rightarrow 6s6p\ ^3\mathrm{P}_1$ transitions designed to induce a specific combination of vector and tensor light shifts [28]. For convenience, the same Rabi beam is also used to drive Rabi transitions in ^{171}Yb, where it induces a pure vector shift in the spin-1/2 system. In this work, the induced Rabi transition frequency is $2\pi \times 0.4$ kHz for ^{173}Yb, and $2\pi \times 3$ kHz for ^{171}Yb.

Ramsey interferometry is simultaneously implemented on both isotopes. Its measurement sequence is illustrated in Fig. 1b. Since all laser pulses are simultaneously shone on atoms of both isotopes, care needs to be taken either to avoid crosstalks between the two isotopes, or to compensate for any crosstalk effects. First, optical pumping at 399 nm along the Z-axis is applied to polarize ^{173}Yb and ^{171}Yb successively. Then, a $\pi/2$-pulse of the 556 nm Rabi beam is applied to drive the spin state of ^{171}Yb. The pulse also induces a small rotation ($\sim \pi/16$) on the spin of ^{173}Yb. After a quarter of the ^{171}Yb-Larmor period ($\mathrm{T}_{171}/4$), a $\pi/2$-pulse of the Rabi beam is applied to drive ^{173}Yb into the cat state, while having little effect on ^{171}Yb because their spins are pointing along the Rabi beam direction at the time of the pulse. Afterwards, spins of both isotopes are allowed to precess freely and accumulate phases over time. At the readout stage, first on ^{173}Yb, the phase information is converted into population values by applying a $\pi/2$-pulse to the precessing cat state. The populations are measured as follows (see Methods): the first probe pulse at 556 nm measures the population of $|5/2, +5/2\rangle_g$, yielding N_{173+}; a subsequent π-pulse exchanges the populations between $|5/2, +5/2\rangle_g$ and $|5/2, -5/2\rangle_g$; the second probe then captures the original population of $|5/2, -5/2\rangle_g$, denoted as N_{173-}. The differential signal is derived as $S_z = (N_+ - N_-)/(N_+ + N_-)$. Note that both the second $\pi/2$-pulse and the π-pulse on ^{173}Yb unavoidably operate on ^{171}Yb as well. To compensate their effects on ^{171}Yb, an additional pulse is introduced so that the three pulses, each successive one delayed by a complete ^{171}Yb-Larmor period (T_{171}), combine to rotate the spin of ^{171}Yb by $(12 + 1/2)\pi$. Finally, the populations of ^{171}Yb spin states are read out in a way similar to that of ^{173}Yb.

The resulting Ramsey fringes are displayed in Fig. 1c. The contrast of ^{173}Yb decays due to interspecies collision between ^{173}Yb and ^{171}Yb, reducing the coherence time of ^{173}Yb from thousands of seconds down to 60 s. Meanwhile, the collisions seem to have no degradation effects on ^{171}Yb.

By applying simultaneous Ramsey interferometry on both isotopes, we determine the precession phases, and derived the Larmor frequencies of ^{171}Yb (ω_{171}) as well as the cat state of ^{173}Yb (ω_{173}), both with a statistical sensitivity of $\sim\ 2\pi \times 2$ mHz in a single-shot measurement.

Under ideal conditions, the comagnetometer frequency ratio $R_L = \omega_{171}/\omega_{173}$ is insensitive to magnetic-field fluctuation. However, we find that light shifts induced by the high-power optical lattice could cause a significant perturbation to R_L.

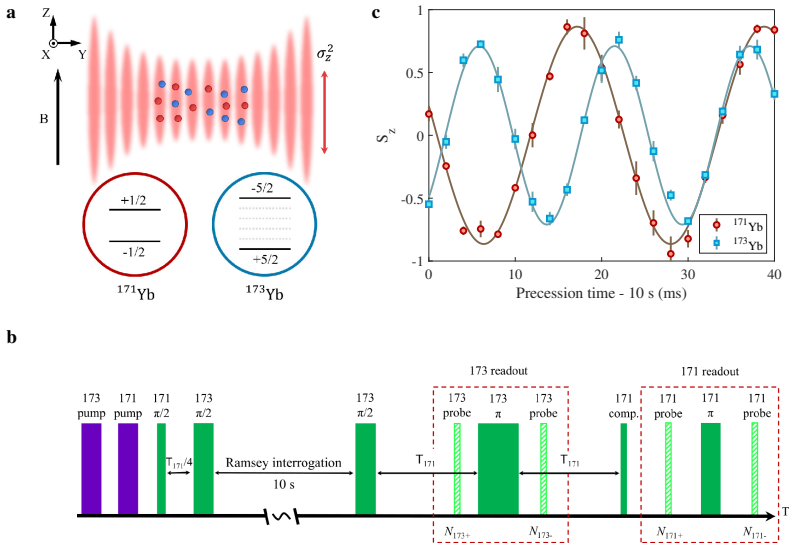

Fig. 1 ^{171}Yb-^{173}Yb cold-atom comagnetometer. (a) Atoms of ^{171}Yb and ^{173}Yb are mixed and confined together in the optical lattice. A uniform field of 10-100 mG is applied to define the quantization axis in the vertical Z-direction. While ^{171}Yb (I = 1/2) naturally has no tensor light shifts, ^{173}Yb (I = 5/2) is prepared into the cat state, which is immune to decoherence caused by the tensor shifts in the optical lattice. (b) Measurement sequence of the simultaneous Ramsey interferometry. The arrangement of the 399 nm pump pulses (solid purple), 556 nm Rabi pulses (solid dark green), and 556 nm probe pulses (hatched light green) are explained in details in the main text. (c) Ramsey signals at 10 s with a bias field at 50 mG, where red circles represent the precession signals of ^{171}Yb, and blue squares correspond to those of ^{173}Yb. Error bars: one standard deviation (1-σ) derived from 3 repeated measurements; solid curves: least-squares fittings with sinusoidal functions.

Dependence on light shift

The light shift can be decomposed into scalar, vector, and tensor terms, which are respectively proportional to the zeroth, first, and second powers of the magnetic quantum number m_F [29]. The scalar shift causes identical displacements for all m_F states, thereby contributing no phase shifts in the evolution of angular momentum states. The vector shift, induced by circularly polarized light, acts as a Zeeman-like shift with its quantization axis along the wave vector of the light field. Given the linearly polarized optical lattice used in this

work, the tensor shift aligned along the polarization direction emerges as the predominant effect.

Tensor shift

Tensor shifts are naturally absent in the ground state $^1S_0$ of ^{171}Yb (I = 1/2). For ^{173}Yb (I = 5/2), tensor shifts are also expected to be null in the cat state of $^1S_0$, where perfect alignment between the lattice polarization and the magnetic field B would make the alignment state an eigenstate of the tensor shift. However, if a small angle θ exists between the polarization (z) and B (along Z), the misalignment twists the alignment state, producing a frequency shift through a mechanism similar to that described by alignment-to-orientation conversion (AOC) [30]. ~~but instead are found in this work to affect the Larmor frequency in a significant and peculiar way. The tensor operator aligns along the linear polarization direction (z) of the lattice, which, due to imperfection, form a small angle θ with the bias magnetic field B in the Z-direction.~~ Without loss of generality, assume the magnetic field has a small component in the transverse x-direction, the Hamiltonian $\mathscr{H}$ arising from the combined magnetic and tensor fields can be expressed as

$$\mathscr{H} = \Omega_{\mathrm{B}}(\sigma_z + \theta\sigma_x) + \Omega_{\mathrm{T}}\sigma_z^2 \ (\theta \ll 1). \tag{1}$$

Here, σ_z and σ_x are the angular momentum matrices, with their terms corresponding to Zeeman effects. The σ_z^2 term describes the tensor effect. Ω_{B} and Ω_{T} are the corresponding coupling parameters.

Under constant optical trapping conditions, we scan the orientation of the bias magnetic field over two dimensions and map the Larmor-frequency ratio ~~comagnetometer ratio~~ R_L. First, the magnetic field is set at 50 mG, corresponding to $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 2.2$ (see Methods). The measured R_L values fall onto a parabolic surface with a minimum extremum $R_{L,min} = 0.726079(3)$ (Fig. 2b). Next, at a higher magnetic field of 100 mG, where $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 4.3$, the opening direction of the parabolic surface is found to be inverted, yielding a maximum $R_{L,max} = 0.726080(5)$ (Fig. 2d), with its value consistent with the minimum value obtained at 50 mG. The parabolic surface and its curvature inversion can be attributed to a second-order perturbation effect on the ^{173}Yb energy levels, with the transverse σ_x term treated as a perturbation. The resulting shifts on the energy splitting between $|5/2, +5/2\rangle_g$ and $|5/2, -5/2\rangle_g$ can be expressed as

$$\Delta E := \delta E^{(2)}_{+5/2} - \delta E^{(2)}_{-5/2} = \frac{5}{2} \cdot \left(\frac{\Omega_{\mathrm{B}}}{\Omega_{\mathrm{T}}}\right)^2 \cdot \frac{\Omega_{\mathrm{B}}\theta^2}{(\frac{\Omega_{\mathrm{B}}}{\Omega_{\mathrm{T}}} + 4)(\frac{\Omega_{\mathrm{B}}}{\Omega_{\mathrm{T}}} - 4)}, \tag{2}$$

where $\delta E^{(2)}_{m_F}$ denotes the 2nd-order perturbation correction to the energy level of $|5/2, m_F\rangle_g$. Derivation of Eq. (2) is presented in Methods. It can be seen that, on opposite sides of the critical ratio $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 4$, the sign of ΔE reverses, resulting in curvature inversion of the R_L surface. At the critical ratio of $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 4$, due to the degeneracy between $|5/2, -5/2\rangle_g$ and $|5/2, -3/2\rangle_g$,

the above perturbation treatment is no longer valid, and the cat state can no longer be preserved.

In order to evaluate the robustness of the comagnetometer against environmental magnetic field fluctuations, we define the suppression factor η as:

$$\eta(\mathrm{B}) = \left| \frac{\Delta \mathrm{B}}{\mathrm{B}} \Big/ \frac{\Delta R_L}{R_L} \right|. \tag{3}$$

From 50 mG to 100 mG, the magnetic field changes by a factor of two, so are the Larmor frequencies of each isotope, yet the Larmor-frequency ratio ~~comagnetometer ratio~~ R_L remains constant at the 10 ppm level. We derive the suppression factor $\eta > 3 \times 10^4$ (95% confidence level) based on the two R_L values and their corresponding uncertainties.

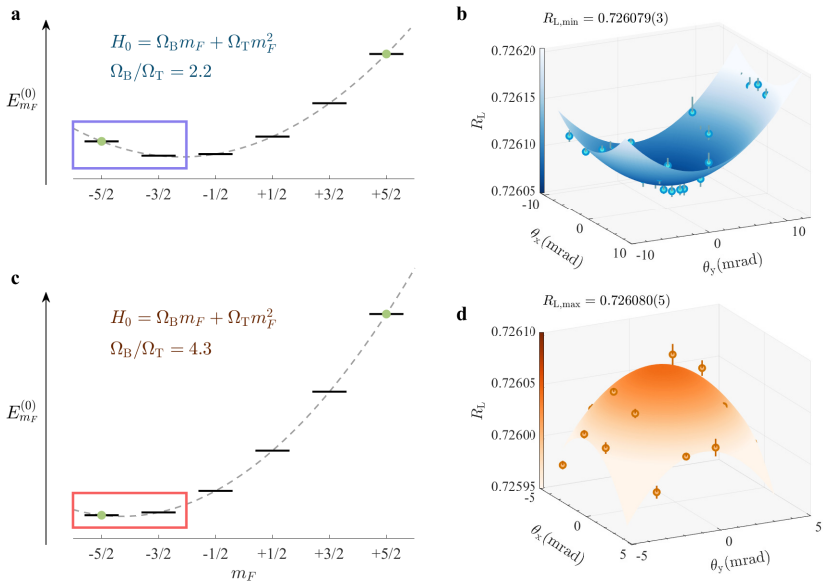

Fig. 2 The combined effect of Zeeman and tensor shifts on the Larmor frequency (ω_{173}) of the cat state of ^{173}Yb, and on the Larmor-frequency ratio ~~comagnetometer ratio~~ $R_L = \omega_{171}/\omega_{173}$. ^{173}Yb atoms are populated in $|5/2, \pm 5/2\rangle_g$ (green dots). (a) At $\Omega_\mathrm{B}/\Omega_\mathrm{T} = 2.2$, where $|5/2, -5/2\rangle_g$ is shifted higher than $|5/2, -3/2\rangle_g$, the σ_x term mixes the two states in the rectangle and pushes $|5/2, -5/2\rangle_g$ further up. This results in a lower frequency ω_{173}, and a higher R_L value. (b) Under 50 mG, corresponding to $\Omega_\mathrm{B}/\Omega_\mathrm{T} = 2.2$, R_L is mapped as the magnetic-field orientation is scanned around the Z-direction. The higher R_L values in (a) results in a surface with positive curvatures. (c) At $\Omega_\mathrm{B}/\Omega_\mathrm{T} = 4.3$, where $|5/2, -5/2\rangle_g$ is shifted below $|5/2, -3/2\rangle_g$, the σ_x term pushes $|5/2, -5/2\rangle_g$ further down. (d) Under 100 mG, corresponding to $\Omega_\mathrm{B}/\Omega_\mathrm{T} = 4.3$. The curvature of the surface is found to reverse. Error bars: one standard deviation (1-σ) derived from 50 repeated measurements; surfaces: least-squares fittings with quadratic surface functions.

Residual vector shift

In order to minimize vector shifts, the optical lattice is tuned to be as close to linearly polarized as possible. This is done by optimizing the coherence of the Ramsey interferometry of ^{171}Yb. During polarization tuning, the bias magnetic field is temporarily redirected to the horizontal propagation direction of the optical lattice (Y-axis), and its strength lowered to 1 mG, both measures taken to deliberately enhance the decoherence effect due to the inhomogeneous vector shifts in the lattice. In this way, the residual circular polarization component, as measured by the Stokes parameter S_3, is tuned to be less than 0.1%. According to calculated vector polarizabilities of ^{171}Yb and ^{173}Yb, the vector shifts due to the residue circular polarization can be regarded as effective magnetic fields less than 100 μG along the Y-direction. At a bias field of 50 mG in the Z-direction, the relative deviation in R_L due to the residue vector shift is estimated to be less than 3×10^{-7}, which is negligible compared to the current measurement uncertainty of 3×10^{-6}. When the field is increased to 100 mG, the deviation drops further down to 8×10^{-8}.

Ratio of nuclear magnetic moments

We have investigated systematic uncertainties in the measurements of the Larmor-frequency ratio ~~comagnetometer ratio~~ R_L. After aligning the 100 mG magnetic field to be parallel to the polarization direction of the optical lattice, we measure R_L while varying the lattice power by more than a factor of five. No dependence of R_L on light power is observed within the measurement uncertainties (Fig. 3a). A weighted average across different lattice powers yields $R_L = 0.726077(2)$. Separately, we measure R_L while varying the density ratio of ^{171}Yb over ^{173}Yb by a factor of seven. This is to investigate possible collisional shifts that are likely to be density dependent. Again, no dependence of R_L on the density ratio is observed (Fig. 3b). The average value across different density ratios is 0.726072(4). The Earth's rotation frequency of $2\pi \times 11.6$ μHz is negligible compared to the measurement uncertainty of around $2\pi \times 100$ μHz for the Larmor frequencies.

Since the two aforementioned independently determined values of R_L agree, we take a weighted average of both, ~~We calculate a weighted average of comagnetometer ratio R_L across different experimental parameters,~~ yielding a final value of $R_L = 0.726076(3)$ and the ratio of nuclear magnetic moments between ^{171}Yb and ^{173}Yb, $R_{NMM} = \mu_{I,171}/\mu_{I,173} = -0.726076(3)$. As shown in Table 1, the new result is consistent with the reported values from prior work [31–33], while improving the precision by an order of magnitude. ~~This new result agrees with the recommended value of −0.72610(4)[31–33] while improving the precision by an order of magnitude.~~

Table 1 Compilation of reported values. Error bars: one standard deviation (1-σ).

Reference	$\mathbf{R_{NMM}}$
Gossard, A. C. et al., 1964 [31]	-0.7278(16)
Olschewski, L., 1972 [32]	-0.72610(4)
This work	-0.726076(3)

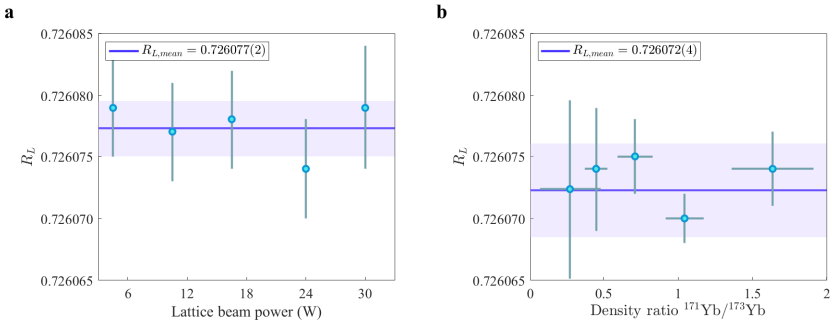

Fig. 3 Investigating possible systematic effects. (a) In a magnetic field of 100 mG, the lattice beam power is varied. (b) With both the magnetic field and lattice parameters fixed, the density ratio between ^{171}Yb and ^{173}Yb is varied. No dependences on these experimental parameters are observed. Error bars: one standard deviation (1-σ) derived from 50 repeated measurements; solid lines: weighted averages.

Discussion

With its strong suppression of magnetic-field noise and tight atom localization within the lattice sites, our cold-atom comagnetometer enables sensitive searches for exotic interactions beyond the Standard Model, particularly at short interaction ranges of 1-100 μm. Among many new-physics interactions proposed [3, 34], we present two examples highlighting the capability of this system.

The comagnetometer is used to search for P- and T-violating spin-mass couplings, also known as monopole-dipole interactions [12, 13, 19, 35]. In this search, with ^{171}Yb and ^{173}Yb atoms as spin sources, the Larmor-frequency ratio ~~comagnetometer ratio~~ is monitored as a dense mass source is brought into proximity with the atoms. By using optical tweezers to position the atoms, the atom-mass separation can be precisely controlled down to a few micrometers. We estimate that, with an interrogation time of 10 s per measurement, a single-atom projection noise, a minimum atom-mass separation of 5 μm, and 30 days of data accumulation, this setup can improve the sensitivity to spin-mass couplings at a 1 μm interaction range by up to three orders of magnitude compared to the current best experimental limits.[36] ~~We estimate that this setup can improve the sensitivity to spin-mass couplings at the 1 μm~~

~~interaction range by up to three orders of magnitude compared to the current experimental limits [36].~~

Furthermore, we aim to improve the search sensitivity of electric dipole moments (EDMs) using cold atoms [5, 37]. The current EDM experiments on ^{171}Yb [5] are limited by magnetic-field noises, particularly the Johnson noise in adjacent electrodes. A ^{171}Yb-^{173}Yb cold-atom comagnetometer can effectively suppress such noise effects, thereby boosting the EDM search sensitivity. ~~Furthermore, EDM searches have been performed on cold atoms of ^{225}Ra (I = 1/2), an isotope whose octupole-deformed nucleus provides enhanced sensitivity to T-violating interactions [37]. As a future direction, we propose forming a comagnetometer using ^{223}Ra (I = 3/2) and ^{225}Ra to suppress magnetic noise in the next-generation ^{225}Ra EDM experiments.~~

Methods

Zeeman and tensor light shifts of ^{173}Yb

The Hamiltonian in Eq. (1) is rewritten below, treating the σ_x term as a perturbation:

$$\begin{aligned}\mathscr{H} &= \mathscr{H}_0 + \theta\mathscr{H}' \\ &= \Omega_{\mathrm{B}}\sigma_z + \Omega_{\mathrm{T}}\sigma_z^2 + \theta\Omega_{\mathrm{B}}\sigma_x \ (\theta \ll 1).\end{aligned} \tag{4}$$

The perturbed eigenenergies are

$$E_{m_F} = E^{(0)}_{m_F} + \theta \left\langle \psi^{(0)}_{m_F} \middle| \mathscr{H}' \middle| \psi^{(0)}_{m_F} \right\rangle + \sum_{i \neq m_F} \theta^2 \frac{\left\langle \psi^{(0)}_i \middle| \mathscr{H}' \middle| \psi^{(0)}_{m_F} \right\rangle^2}{E^{(0)}_{m_F} - E^{(0)}_i} + O(\theta^3), \tag{5}$$

$$E^{(0)}_{m_F} = m_F\Omega_{\mathrm{B}} + m_F^2\Omega_{\mathrm{T}}, \tag{6}$$

where the θ^2 term corresponds to $\delta E^{(2)}_{m_F}$ in Eq. (2). When $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 4$, the unperturbed energy levels of $|5/2, -5/2\rangle_g$ and $|5/2, -3/2\rangle_g$ are degenerate.

Here the first-order perturbation term is always zero because the σ_x term corresponds to a subdiagonal matrix. For the spin-5/2 system in ^{173}Yb, the σ_x matrix is

$$\sigma_x = \begin{pmatrix} 0 & \frac{\sqrt{5}}{2} & & & & \\ \frac{\sqrt{5}}{2} & 0 & \frac{\sqrt{8}}{2} & & & \\ & \frac{\sqrt{8}}{2} & 0 & \frac{\sqrt{9}}{2} & & \\ & & \frac{\sqrt{9}}{2} & 0 & \frac{\sqrt{8}}{2} & \\ & & & \frac{\sqrt{8}}{2} & 0 & \frac{\sqrt{5}}{2} \\ & & & & \frac{\sqrt{5}}{2} & 0 \end{pmatrix}. \tag{7}$$

To calculate the Larmor frequency of the cat state of ^{173}Yb, we consider the 2nd-order perturbation on the two populated stretched states $|5/2, \pm 5/2\rangle_g$:

$$\delta E^{(2)}_{+5/2} = \theta^2 \cdot \frac{5}{4} \cdot \frac{\Omega_{\mathrm{B}}^2}{\Omega_{\mathrm{B}} + 4\Omega_{\mathrm{T}}}, \tag{8}$$

$$\delta E^{(2)}_{-5/2} = \theta^2 \cdot \frac{5}{4} \cdot \frac{\Omega_{\mathrm{B}}^2}{-\Omega_{\mathrm{B}} + 4\Omega_{\mathrm{T}}}. \tag{9}$$

Subtracting the two energy shifts yields Eq. (2).

Calibration of $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}}$

We perform precession measurements on ^{173}Yb to calibrate the $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}}$ values. Since the cat state is immune to tensor shifts (Ω_{T}), we employ a coherent spin state (CSS) encompasses all m_F substates instead, whose spin is fully polarized along a direction.

While keeping the lattice linearly polarized along the z-axis, its power is reduced to 8 W to mitigate the otherwise rapid decoherence induced by the tensor shift. Meanwhile, a bias magnetic field of 23.5 mG is applied along the x-direction. The ^{173}Yb atoms are first pumped into $|5/2, +5/2\rangle_g$, then allowed to precess under the influence of both the Zeeman and tensor shifts, $\mathscr{H}_{calib} = \Omega_{\mathrm{B}}\sigma_x + \Omega_{\mathrm{T}}\sigma_z^2$. After a variable precession duration (0 – 2,500 ms), the population signal S_z is measured along the z-axis (Fig. 4). The detection probes the ensemble collectively, and the signal comprises contributions from all m_F substates.

We simulate the precession process of ^{173}Yb using the calibration Hamiltonian ($\mathscr{H}_{calib}$) and compare the results with experimental data. ~~, obtaining $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 3.5$ under the given magnetic field and light intensity conditions~~ Our simulation models the observed population signal as a sum over all m_F states. The contribution of each state is weighted by the Clebsch-Gordan coefficients of the relevant transitions and the detuning of the probe beam. Specifically, when probing $|5/2, +5/2\rangle_g$, the detected signal comprises contributions from all $|5/2, m_F\rangle_g$ states (m_F = +5/2 to -5/2), with relative weights of 1, 0.58, 0.31, 0.16, 0.08, and 0.03. Furthermore, the spatial distribution of the tensor field is incorporated via a Monte Carlo method. The resulting theoretical curve shape is highly sensitive to the ratio $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}}$. This sensitivity allows us to extract $\Omega_{\mathrm{B}}/\Omega_{\mathrm{T}} = 3.5$ from the good agreement between the simulation and the experimental data under the given magnetic field and lattice intensity conditions.

It is worth noting that the detection issue described here is specific to CSS used for calibration. In contrast, the cat states prepared elsewhere in the experiment involve only $|5/2, \pm 5/2\rangle_g$ sublevels, exhibiting the maximally different detection probabilities, thus being clearly resolvable. Details are provided in the Methods section "Atom sample detections" below.

Atom sample detection

A circularly polarized laser probe beam along the Z-direction runs through the cold atom ensemble and is detected by a CMOS camera. Absorption images are

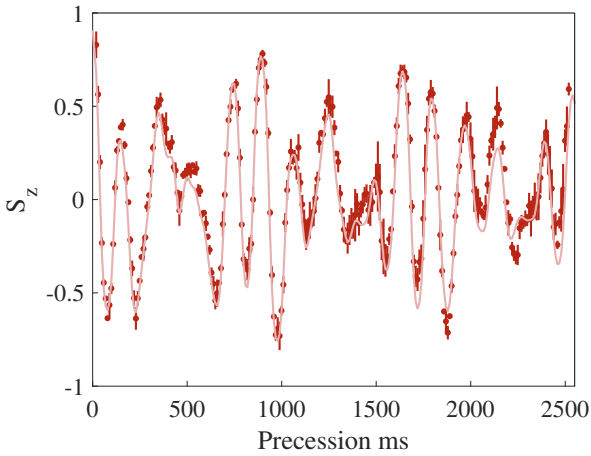

Fig. 4 Precession of a coherent spin state of ^{173}Yb under $B_x = 23.5$ mG and 8 W lattice power. Error bars: one standard deviation (1-σ) derived from 3 repeated measurements. The solid curve represents simulation results.

taken on the atoms for population measurements. The state readout of atoms is performed via differential detection of two stretched states. For ^{171}Yb atoms, the probe beam resonantly excites the cycling transition $|1/2, +1/2\rangle_g \rightarrow |3/2, +3/2\rangle_e$ (Fig. 5a). Meanwhile, the $|1/2, -1/2\rangle_g \rightarrow |3/2, +1/2\rangle_e$ transition is off resonance by 10 times the natural linewidth (10 Γ) due to tensor shifts on the excited states. As a result, the cycling transition can be repeated 1,200 times on average before a spin flip occurs. This spin selectivity ensures that the spin state is detected with an efficiency approaching 100%. For ^{173}Yb atoms, the relevant transitions are $|5/2, +5/2\rangle_g \rightarrow |7/2, +7/2\rangle_e$ and $|5/2, -5/2\rangle_g \rightarrow |3/2, -3/2\rangle_e$ (Fig. 5b). In addition to a comparable difference of tensor shifts ($\sim$ 20 Γ), the branching ratios of these two transitions differ by a factor of 21, thus further enhancing the spin selectivity.

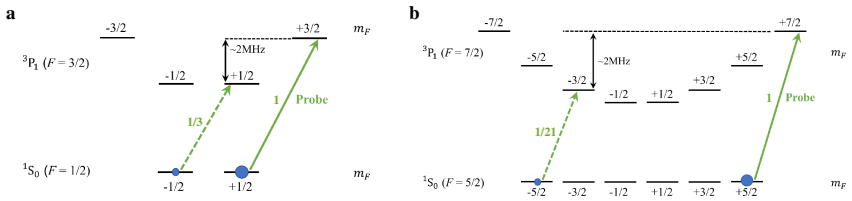

Fig. 5 Transitions used to measure spin-state populations in ^{171}Yb (a) and ^{173}Yb (b). The arrow of solid line indicates a cycling transition used for population measurements; the arrow of dashed line indicates a spin-flip transition suppressed by large detunings.

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Declarations

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- Author contribution
  J.-L. Z. and W.-T. L. constructed the experimental apparatus and performed the experiments and simulations. Y. A. Y. contributed to the

apparatus during the early stages. Y.-Q. W. made contributions to the theory part. J.-L. Z., W.-T. L., Y.-Q. W., T. X. and Z.-T. L. carried out data analysis and wrote the manuscript.

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